Chapter 4
MATHEMATICS OF CHANCE

1. A basket contains 18 ripen mangoes and some are not ripen. If a mango is taken from basket, without looking it the probability of getting a ripen mango is $\frac{3}{5}$. Then the total number of not ripen mangoes is?

   A) 20 B) 10

   C) 12 D) 16

2. In a leap year, probability of 53 Friday is,

   A) $\frac{3}{7}$ B) $\frac{2}{7}$

   C) $\frac{4}{7}$ D) $\frac{5}{7}$

3. Two dice each marked with numbers 1 to 6 are rolled together. What is the probability of getting both have same number?

   A) $\frac{1}{6}$ B) $\frac{2}{6}$

   C) $\frac{3}{6}$ D) $\frac{5}{6}$

4. Natural number from 1 to 40 are written on paper slips and are put in a box. One slip is taken from the box. What is the probability of getting a multiple of 5?

   A) $\frac{1}{5}$ B) $\frac{3}{5}$

   C) $\frac{4}{5}$ D) 1

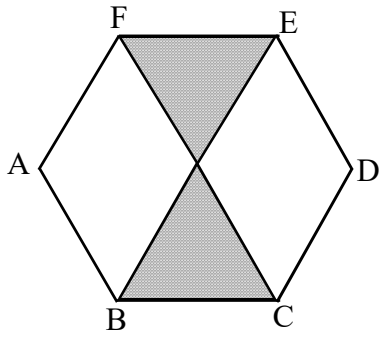
5. A box contains white balls and black balls. Total number of ball is 18. If a ball is taken from it, probability of it being black is $\frac{1}{3}$, then what is the number of white balls?

   A) 3 B) 9

   C) 6 D) 12

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6.



Statement 1: In the figure, ABCDEF is regular polygon. A dot is put inside the regular hexagon with out looking it. The probability that the dot to be within the shaded part is $\frac{1}{3}$.

Statement 2 : In an equilateral triangle, angles are 60^0

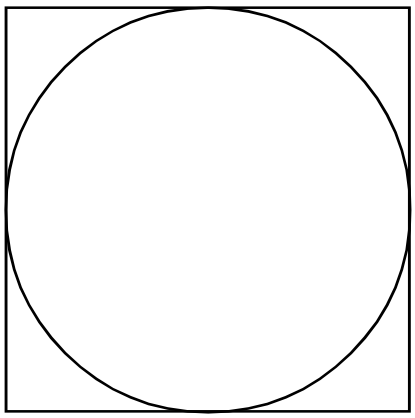
A) Statement 1 is true, statement 2 is false

B) Statemet 2 is true, statement 1 is false

C) Both statements are true, statement 2 is the correct reason of statement 1

D) Both statements are true, statement 2 is not the correct reason of statement 1

7.



Statement 1 : A circle is drawn inside the circle. A dot put inside the square with out looking it, the probability that the dot to be withing the circle is $\frac{\pi}{4}$.

Statement 2 : The diameter of a circle is 2cm. Its area is 4π square c.m.

Choose the correct answer from the following.

A) Statement 1 is true, statement 2 is false

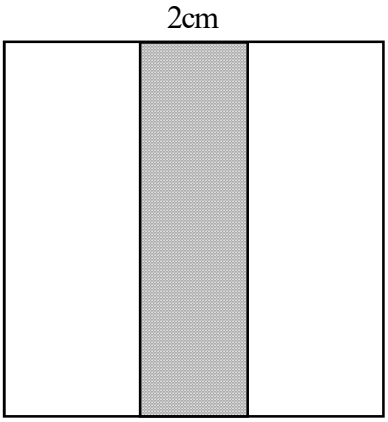
B) Statemet 2 is true, statement 1 is false

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C) Both statements are true, statement 2 is the correct reason of statement 1

D) Both statements are true, statement 2 is not the correct reason of statement 1

8.



In the figure, a rectangular portion is shaded in the square of side 5cm.

Statement 1 : A dot is put inside the square without looking it, the probability that the dot lies on the shaded part and unshaded part are equal.

Statement 2 : A dot is put inside the square without looking it, the sum of the probabilities of dot being on the shaded part and unshaded part is equal to 1.

A) Statement 1 is true, statement 2 is false

B) Statemet 2 is true, statement 1 is false

C) Both statements are true, statement 2 is the correct reason of statement 1

D) Both statements are true, statement 2 is not the correct reason of statement 1

9. Statement 1 : One die with numbers 1 to 6 are rolled, then the probability of getting prime number is $\frac{1}{2}$.

Statement 2 : Prime numbers on a dice are 2, 3 and 5.

A) Statement 1 is true, statement 2 is false

B) Statemet 2 is true, statement 1 is false

C) Both statements are true, statement 2 is the correct reason of statement 1

D) Both statements are true, statement 2 is not the correct reason of statement 1

10. Statement 1 : Natural numbers from 1 to 15 are written on paper slips and are put in a box. One slip is taken from the box with out looking it, then probability of even number is $\frac{7}{15}$.

Statement 2 : Total number of even numbers between 1 to 15 is 7.

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A) Statement 1 is true, statement 2 is false

B) Statemet 2 is true, statement 1 is false

C) Both statements are true, statement 2 is the correct reason of statement 1

D) Both statements are true, statement 2 is not the correct reason of statement 1

11. One asked to say a two digit number, what is the probability that the number is a perfect square?

12. Natural numbers from 1 to 30 are written on paper slips and are put in a box. One slip is taken from each box without looking.

    i) What is the probability of it being an even number?

    ii) What is the probability of it being a prime number?

13. A box contains 15 red beads and some blue beads. It we take one bead from this box without looking, then probability of getting blue bead is $\frac{4}{9}$.

    i) What is the probability of getting red bead?

    ii) Find the number of blue beads?

    iii) If we add 3 more red bead, then what is the probability of getting blue bead?

14. In a class, the total number of students is 40. One student is to be selected from the class. The probability of selecting a boy from the class is $\frac{3}{8}$.

    i) How many boys are there in the class?

    ii) How many girls are there in the class?

    iii) How many more boys should be join to the class so that the probabiity of getting a boy is $\frac{1}{2}$.

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15. One is asked to say a two digit number

    i) Find the total number of two digit number.

    ii) What is the probability of both digits being the same.

    iii) What is the probability of the first digit being greater than the second?

    iv) What is the probability of the first digit being less than the second?

16. Two boxes containing natural numbers from 1 to 10. One slip is taken from each box, without looking. What is the probability that the number on the both slips being a prime number?

17. A bag contains 5 red balls and some blue balls. One ball is taken from each bag without looking. The probability of getting a blue ball is three times the probability of getting a red ball.

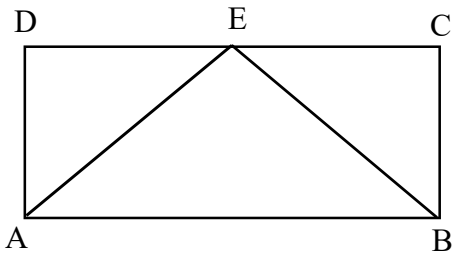
    i) If the total number of blue ball is x, then find the total number of balls.

    ii) What is the probability of getting red ball?

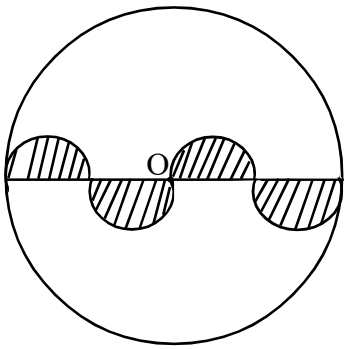
    iii) What is the probability of getting blue ball?

    iv) Find the total number of blue balls?

18. ABCD is a rectangle; E is the mid-point of CD. If we put a dot in the figure without looking it. What is the probability that it would be outside triangle AEB?



19. In the figure, all semicircles have equal areas. A dot is put inside the circle with out looking it, what is the probability that the dot lies on the shaded part



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20. One is asked to say a natural numbers from 5 to 30.

i) What is the probability of it being an even number?

ii) What is the probability of it being a prime number?

21. A person is asked to say a three-digit number.

i) What is the probability that all the three digits of this number are same?

ii) What is the probability that the number is an palindrome number?

22. There are 20 boys and 25 girls in Class 10 A and 30 boys and 20 girls in Class 10B. One student is to be chosen from each class.

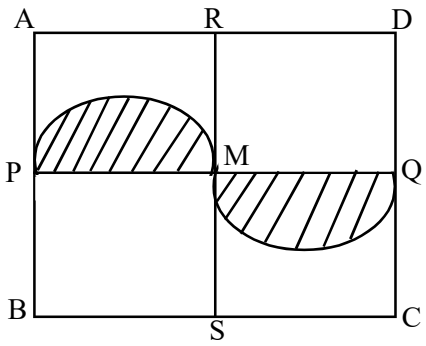
i) What is the total number of possible selections?

ii) What is the probability of both being boys?

iii) What is the probability of one being a boy and one being a girl?

iv) What is the probability of at least one being a girl?

23. In the figure, PQ, RS are the lines joining the midpoints of the sides of square ABCD. Two semicircles are drawn with PM and QM as diameters. A dot is marked inside the square with out looking it. What is the probability that the dot lies on the shaded part.



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24. Two dice with colour red and black, each marked with numbers 1 to 6 are roled together.

i) If the possible numbers got from dice are written as pairs, how many pairs would be there?

ii) In how many pairs, the product of the numbers is odd?

iii) What is the probability of getting sum odd?

iv) What is the probability of getting an odd number on the red die and even number on the black die?

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Chapter 4

MATHEMATICS OF CHANCE

ANSWERS

1. C) 12
2. B) $\frac{2}{7}$
3. A) $\frac{1}{6}$
4. A) $\frac{1}{5}$
5. D) 12
6. C) Both statements are true, statement 2 is the correct reason of statement 1
7. A) Statement 1 is true, statement 2 is false
8. B) Statemet 2 is true, statement 1 is false
9. C) Both statements are true, statement 2 is the correct reason of statement 1
10. C) Both statements are true, statement 2 is the correct reason of statement 1
11. Perfect squares

    1, 4, 9, ... 81, 100

    Two digit perfect sqaures

    16, 25, 36, 49, 64, 81

    Probability = $\frac{6}{90}$

12. i) $\frac{15}{30}$

    ii) Prime numbers

    2, 3, 5, 11, 13, 17, 19, 23, 29

    Chance = $\frac{10}{30}$

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13. blue beads $= \frac{4}{9}$

i) red beads = $\frac{5}{9}$

ii) $\frac{5}{9}x = 15$

$x = \frac{15x9}{5} = 27$

blue beads = 27-15 = 12

iii) Chance = $\frac{12}{30}$

14. i) $40x\frac{3}{8} = 15$

ii) $40 - 15 = 25$

iii) 10

15. i) 90

ii) $\frac{9}{90}$

iii) $\frac{45}{90}$

iv) $\frac{36}{90}$

16. $\frac{16}{100}$

17. i) $5 + x$

ii) $\frac{5}{5+x}$

iii) $\frac{x}{5+x}$

iv) $\frac{3x5}{5+x} = \frac{x}{x+5}$

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$x = 15$

No.of blue balls = 15

18. $\frac{1}{2}$

19. Consider radius of by circle 2r,

Radius of small circle is = $\frac{r}{2}$

Area of 4 semi circles = $\frac{4x\pi r^2}{8} = \frac{\pi r^2}{2}$

Area of big circles = $4\pi r^2$

Probability = $\frac{\pi r^2}{2} \div 4\pi r^2$

$= \frac{1}{8}$

20. Even numbers = 6, 8, ... 30

i) $\frac{13}{26}$

ii) Prime numbers 5, 7, 11, 13, 17, 19, 23, 29

$\frac{8}{26}$

21. i) $\frac{9}{900}$

ii)

101	201	909	} 10
111	212	919	
121	222	.....	
131	232	.....	
.....	.....	.....	
.....	.....	.....	
191	292	999	

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$$10 \text{ x } 9 = \frac{90}{900}$$

22.

10A	10B
20B	30B
25G	20G
45	50

i) $45x50 = 2250$

ii) $P(BB) = \frac{20x30}{2250} = \frac{600}{2250}$

iii) $P(BG \text{ or } GB) = \frac{20x20 + 25x30}{2250} = \frac{1150}{2250}$

iv) $1 - \frac{600}{2250} = \frac{1650}{2250}$

23. Sides of squares = 4r

Area = $(4r)^2 = 16r^2$

Area of shaded region = πr^2

Probability = $\frac{\pi r^2}{16r^2} = \frac{\pi}{16}$

24. i) 36

ii) $P(OO) = \frac{3x3}{36} = \frac{9}{36}$

iii) $P(OE \text{ or } EO) = \frac{3x3 + 3x3}{36} = \frac{18}{36}$

iv) $P(OE) = \frac{3x3}{36} = \frac{9}{36}$

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